inclass-L35

Thursday, October 22, 2026 8:12 AM

inclass-L35

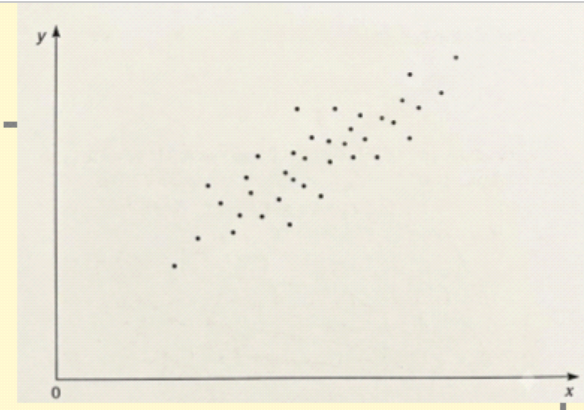
CV2090
Probability & Statistics for Civil Engineers
(Jul – Nov 2026)

Dr. Prakash S Badal

Announcement

- Class/Tutorial rescheduling
  - No Class on 19th Oct
  - No tutorial on 19th Oct

Earlier we saw



Given: (x_i, y_i)

essentially, a set of paired observations

We can "fit" *a* best line

$$y' = \alpha + \beta x$$

We defined this *best* on the basis of least-squared error:

$$\partial\Delta^2/\partial\alpha = 0 \quad \text{and} \quad \partial\Delta^2/\partial\beta = 0$$

Thus, we obtained $\hat{\beta}_{\text{OLS}} = \frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2}$ and $\hat{\alpha}_{\text{OLS}} = \bar{y} - \hat{\beta}_{\text{OLS}}\bar{x}$

And the regression of Y on X becomes: $\hat{y} = \hat{\alpha} + \hat{\beta}x$

Assuming conditional variance to be constant with x

$$s^2_{Y|x} = \frac{\Delta^2}{n-2}$$

"Homoscedastic"

Residual: $y_i - \hat{y}_i$

Δ^2: total cumulative squared error

$s_{Y|x}$: conditional stdev

Note on variance

For Homoscedastic data, i.e., if conditional variance is constant with x

$$s^2_{Y|x} = \frac{\Delta^2}{n-2}$$

What is the unconditional variance of Y?

$$s^2_Y = \frac{\sum(y_i - \bar{y})^2}{n-1}$$

Which one is smaller? Why?

Recall, $\Delta^2 = \sum(y_i - \hat{y}_i)^2$

$$\Delta^2 = \sum(y_i - \hat{y}_i)^2 = \sum[(y_i - \bar{y}) - (\hat{y}_i - \bar{y})]^2$$

$$= \sum\left[(y_i - \bar{y}) - \left(\hat{\beta}\hat{x}_i - \hat{\beta}\bar{x}\right)\right]^2$$

$$= \sum(y_i - \bar{y})^2 + \hat{\beta}^2 \sum(x_i - \bar{x})^2 - 2\hat{\beta} \sum(x_i - \bar{x})\,(y_i - \bar{y})$$

$$= \sum(y_i - \bar{y})^2 - \hat{\beta}^2 \sum(x_i - \bar{x})^2$$

$$\hat{\beta}_{\text{OLS}} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

$s^2_{Y|x} < s^2_Y$ is smaller. What does that mean?

Note on variance

Conditional variance of Y $\quad s^2_{Y|x} = \frac{\Delta^2}{n-2} = \frac{\sum(y_i - \widehat{y_i})^2}{n-2}$

Unconditional variance of Y $\quad s^2_Y = \frac{\sum(y_i - \bar{y})^2}{n-1}$

$$\Delta^2 = \sum(y_i - \bar{y})^2 - \hat{\beta}^2 \sum(x_i - \bar{x})^2$$

Total variation in Y: $\quad \sum(y_i - \bar{y})^2$

Explained variation in Y: $\quad \hat{\beta}^2 \sum(x_i - \bar{x})^2$ (explained by reg.)

Unexplained variation in Y: Total – Explained ≡ Δ^2

'aka' residuals

exactly what we need for the conditional variance $s^2_{Y|x}$

Confidence interval of regression equation

Uncertainty in $\hat{\alpha}$. Constant error

Uncertainty in $\hat{\beta}$. Farther from $\bar{x}$; bigger the error

$$\frac{\overline{Y}_i - \mu_{Y|x_i}}{s_{Y|x}\sqrt{\frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum(x_i - \bar{x})^2}}}$$

follows t-distribution with $(n-2)$ df

The $100(1-\alpha)\%$ CI for the regression equation at several selected values of $X = x_i$ would be:

$$\bar{y}_i \pm t_{(1-\alpha/2),n-2} s_{Y|x} \sqrt{\frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum(x_i - \bar{x})^2}}$$

where

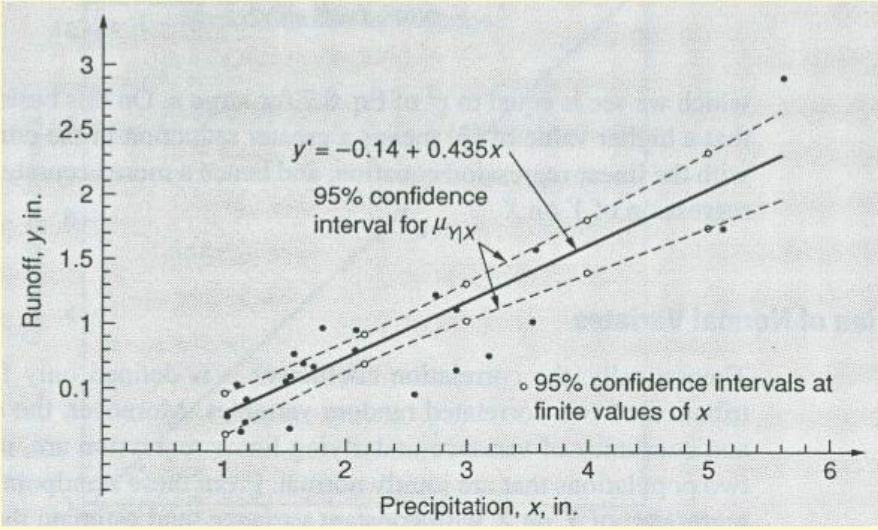
$\bar{y}_i = \mathrm{E}(Y|X = x_i)$;

$s_{Y|x}$ = conditional stdev of Y

Confidence interval of regression equation

The $100(1-\alpha)\%$ CI

$$\bar{y}_i \pm t_{(1-\alpha/2),n-2} S_{Y|x} \sqrt{\frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$$



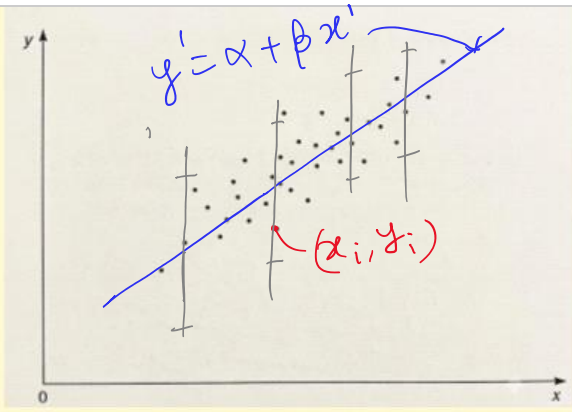
L-35 (3/3; Reg.)
Oct 23, 2026

Earlier we saw

Given: (x_i, y_i)

Best fit line $y' = \alpha + \beta x'$

Residuals: $y_i - y_i'$



Total square error, $\Delta^2 = \sum (y_i - y_i')^2$

Minimize the square error

$\partial \Delta^2 / \partial \alpha = 0$ $\quad \partial \Delta^2 / \partial \beta = 0$

$\hat{\alpha} = \bar{y} - \hat{\beta}\,\bar{x}$ $\quad \hat{\beta}_{OLS} = \dfrac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$

$\hat{\alpha}_{OLS} = \bar{y} - \hat{\beta}_{OLS}\bar{x}$

$\hat{\beta}_{OLS} = \dfrac{\sum (x_i - \bar{x})^2}{\sum (x_i - \bar{x})^2}$

Conditional variance

$s^2_{Y|x} = \Delta^2/(n-2)$

"Homoscedastic"

Heteroscedastic

↳ $s^2_{Y|x}$ depends on x

Statistical Correlation R^2

Ang and Tang (Section 8.2.2, 8.3)

Coefficient of determination, R^2

Generally, $R^2 = [\rho_{(Y, \hat{Y})}]^2$

Specifically, For $\hat{y} = \hat{\alpha} + \hat{\beta} x$ (linear regression with one predictor)

$$R^2 = (\rho_{Y,\hat{Y}})^2 = \left[\frac{COV(Y, \hat{Y})}{\sigma_Y \sigma_{\hat{Y}}}\right]^2 = \frac{\left[\frac{\sum (Y-\bar{Y})(\hat{Y}-\bar{\hat{Y}})}{n-1}\right]^2}{\sigma_Y^2 \quad \hat{\beta}^2 \quad \sigma_X^2}$$

$$= \frac{\beta^2 [COV(Y, X)]^2}{\beta^2 \sigma_Y^2 \sigma_X^2} = \rho_{X,Y}^2$$

$$\Rightarrow \hat{\rho} = \frac{\left[\sum (x_i - \bar{x})(y_i - \bar{y})\right]/n-1}{S_X S_Y} = \hat{\beta} \frac{S_X}{S_Y}$$

$$\Rightarrow \hat{\beta} = \hat{\rho} \frac{S_Y}{S_X}$$

Storm No.	Precip., x_i (in.)	Runoff, y_i (in.)	x_iy_i	x_i^2	y_i^2	$y_i' = \hat{\alpha} + \hat{\beta}x_i$	$(y_i - y_i')^2$
1	1.11	0.52	0.58	1.23	0.270	0.343	0.0313
2	1.17	0.40	0.47	1.37	0.160	0.369	0.0009
3	1.79	0.97	1.74	3.20	0.941	0.637	0.1110
4	5.62	2.92	16.40	31.60	8.526	2.280	0.4000
5	1.13	0.17	0.19	1.28	0.029	0.351	0.0328
6	1.54	0.19	0.29	2.37	0.036	0.530	0.1158
7	3.19	0.76	2.43	10.15	0.578	1.245	0.2360
8	1.73	0.66	1.14	2.99	0.436	0.612	0.0023
9	2.09	0.78	1.63	4.37	0.608	0.770	0.0001
10	2.75	1.24	3.41	7.55	1.538	1.059	0.0328
11	1.20	0.39	0.47	1.44	0.152	0.381	0.0001
12	1.01	0.30	0.30	1.02	0.090	0.299	0.0000
13	1.64	0.70	1.15	2.69	0.490	0.574	0.0158
14	1.57	0.77	1.21	2.46	0.593	0.544	0.0511
15	1.54	0.59	0.91	2.37	0.348	0.530	0.0036
16	2.09	0.95	1.99	4.36	0.902	0.770	0.0326
17	3.54	1.02	3.62	12.55	1.040	1.400	0.1442
18	1.17	0.39	0.46	1.37	0.152	0.368	0.0004
19	1.15	0.23	0.26	1.32	0.053	0.360	0.0169
20	2.57	0.45	1.16	6.60	0.202	0.980	0.2810
21	3.57	1.59	5.66	12.74	2.528	1.415	0.0306
22	5.11	1.74	8.90	26.18	3.028	2.084	0.1185
23	1.52	0.56	0.85	2.31	0.314	0.521	0.0015
24	2.93	1.12	3.28	8.58	1.254	1.135	0.0002
25	2.93	0.64	0.74	1.34	0.410	0.365	0.0755
	$\sum = 53.89$	20.05	59.24	153.44	24.678		$\Delta^2 = 1.7350$

$$\bar{x} = \frac{53.89}{25} = 2.16; \quad \bar{y} = \frac{20.05}{25} = 0.80$$

$$s_X^2 = \frac{1}{24}(153.44 - 25 \times 2.16^2) = 1.533;$$

$$s_Y^2 = \frac{1}{24}(24.68 - 25 \times 0.80^2) = 0.362$$

$$\hat{\beta} = \frac{59.24 - 25 \times 2.16 \times 0.80}{153.44 - 25(2.16)^2} = 0.435;$$

$$\hat{\alpha} = 0.80 - 0.435 \times 2.16 = -0.140$$

$$s_{Y|x}^2 = \frac{\Delta^2}{n-2} = \frac{1.735}{25-2} = 0.075$$

$$\hat{\rho} = \frac{1}{24}\frac{59.24 - 25 \times 2.16 \times 0.80}{\sqrt{1.533}\sqrt{0.362}} = 0.898$$

We can also establish the 95% confidence interval as follows: For this purpose, let us select five discrete values of the precipitation at $x_i = 1.0$, 2.16, 3.0, 4.0, and 5.0 in. Then, according to Eq. 8.8, the respective confidence intervals at these five different values of x_i are, individually, with $t_{0.975,23} = 2.069$ from Table A.3.

$$\text{At } x_i = 1.0: \quad \langle \mu_{Y|1.0} \rangle_{0.95} = 0.295 \pm 2.069 \times 0.274 \sqrt{\frac{1}{25} + \frac{(1.0 - 2.16)^2}{153.44 - 25 \times 2.16^2}}$$
$$= (0.138 \rightarrow 0.452)\,\text{in.}$$

$$\text{At } x_i = 2.16: \quad \langle \mu_{Y|2.16} \rangle_{0.95} = 0.800 \pm 2.069 \times 0.274 \sqrt{\frac{1}{25} + \frac{(2.16 - 2.16)^2}{153.44 - 25 \times 2.16^2}}$$
$$= (0.687 \rightarrow 0.913)\,\text{in.}$$

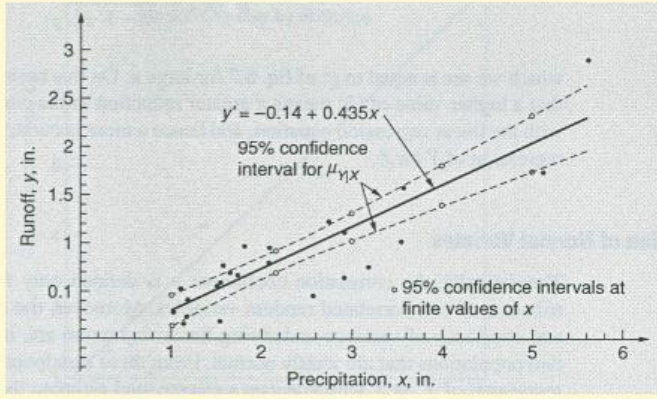
$$\text{At } x_i = 3.0: \quad \langle \mu_{Y|3.0} \rangle_{0.95} = 1.165 \pm 2.069 \times 0.274 \sqrt{\frac{1}{25} + \frac{(3.0 - 2.16)^2}{153.44 - 25 \times 2.16^2}}$$
$$= (1.027 \rightarrow 1.303)\,\text{in.}$$

$$\text{At } x_i = 4.0: \quad \langle \mu_{Y|4.0} \rangle_{0.95} = 1.600 \pm 2.069 \times 0.274 \sqrt{\frac{1}{25} + \frac{(4.0 - 2.16)^2}{153.44 - 25 \times 2.16^2}}$$
$$= (1.368 \rightarrow 1.832)\,\text{in.}$$

$$\text{At } x_i = 5.0: \quad \langle \mu_{Y|5.0} \rangle_{0.95} = 2.035 \pm 2.069 \times 0.274 \sqrt{\frac{1}{25} + \frac{(5.0 - 2.16)^2}{153.44 - 25 \times 2.16^2}}$$
$$= (1.746 \rightarrow 2.324)\,\text{in.}$$

$$E(Y|X = 4) = -0.14 + 0.435 \times 4 = 1.6 \text{ in.}$$

$$P(Y > 2|X = 4) = 1 - \Phi\left(\frac{2 - 1.6}{0.274}\right) = 1 - 0.928 = 0.072$$

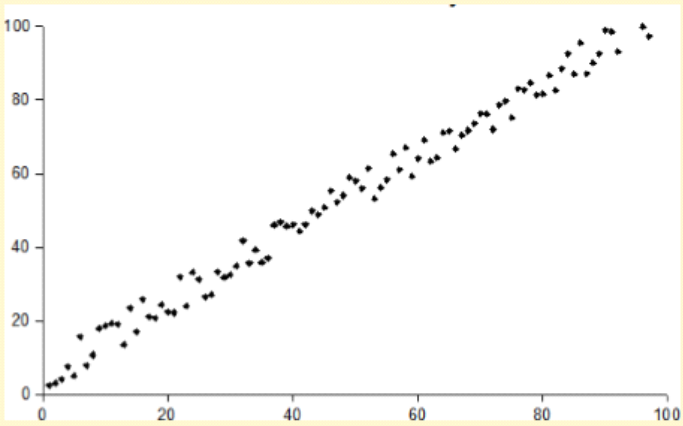


Linear Reg. with Non-constant σ

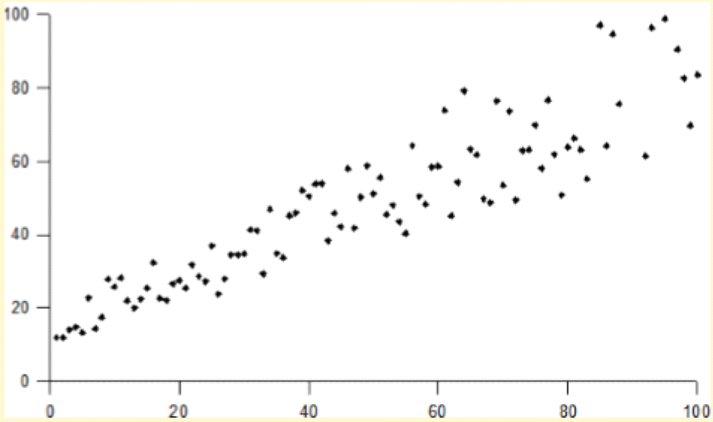
Heteroscedasticity

Ang and Tang (Section 8.4)

Homoscedasticity and Heteroscedasticity



HOMOSCEDASTICITY



HETEROSCEDASTICITY

14

Lin. regression with nonconstant variance

In homoscedastic assumption,

$$s^2_{Y|X} = \frac{\Delta^2}{n-2}$$

To release this assumption,

For heteroscedasticity,

conditional variance

$$s^2_{Y|X} = \sigma^2 g^2(x)$$

where $g(x)$ is a predetermined function.

The regression of Y on X has the same form

$$E[Y|X=x] = \alpha + \beta x$$

α & β: Regression parameters (which may be different than $\hat{\alpha}_{OLS}$ & $\hat{\beta}_{OLS}$)

Reasonable to assume: smaller $s^2_{Y|x}$

$\Rightarrow$ higher weight

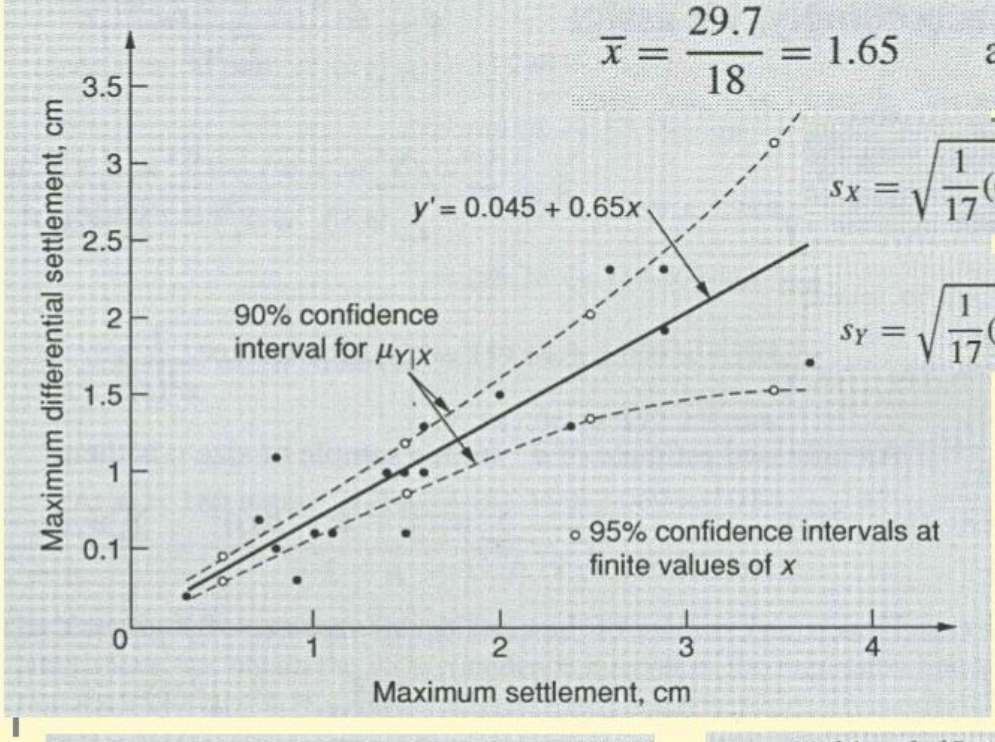
$$w_i' = \frac{1}{s^2_{Y|x_i}} = \frac{1}{\sigma^2 g(x)^2} \Rightarrow \Delta^2_{WLS} = \sum w_i [y_i - (\alpha + \beta x_i)^2]$$

where $w_i = w_i' \sigma^2$

Minimize Δ^2_{WLS} to obtain $\hat{\alpha}_{WLS}$, $\hat{\beta}_{WLS}$ & σ

Lin. regression with nonconstant variance

$$\bar{x} = \frac{29.7}{18} = 1.65 \quad \text{and} \quad \bar{y} = \frac{19.9}{18} = 1.11$$



$$s_X = \sqrt{\frac{1}{17}(63.49 - 18 \times 1.65^2)} = 0.923;$$

$$s_Y = \sqrt{\frac{1}{17}(28.87 - 18 \times 1.11^2)} = 0.627$$

$$\hat{\beta} = \frac{22.38 \times 12.42 - 15.84 \times 11.31}{22.38 \times 18 - 15.84^2} = 0.65$$

$$\hat{\alpha} = \frac{11.31 - 0.65 \times 15.84}{22.38} = 0.045$$

$g(x) = x$, $\text{Var}(Y|X = x) = \sigma^2 x^2$, and $w_i = 1/x_i^2$.

$$E(Y \mid x) = 0.045 + 0.65x.$$

Lin. regression with nonconstant variance

According to Eq. 8.18, the estimate of σ^2 is

$$s^2 = \frac{0.9418}{18-2} = 0.0589$$

and from Eq. 8.19a, we obtain the conditional standard deviation of Y as a function of X to be

$$s_{Y|x} = x\sqrt{0.0589} = 0.243x$$

In this case, the estimated correlation coefficient, from Eq. 8.10, is

$$\hat{\rho} = \hat{\beta}\frac{s_X}{s_Y} = 0.65 \times \frac{0.923}{0.627} = 0.96$$

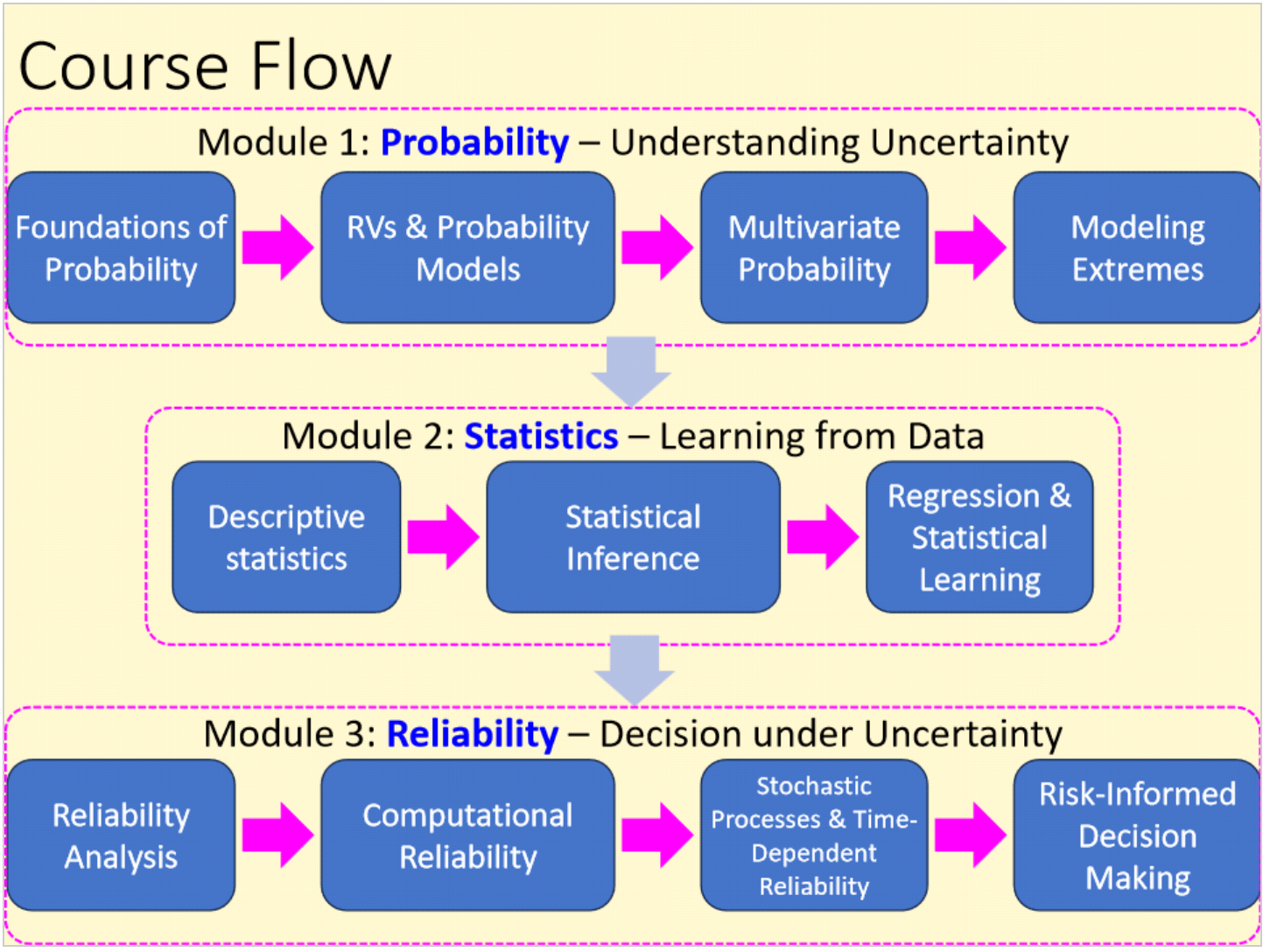
$$\text{At } x_i = 0.5: \quad \langle \mu_{Y|x} \rangle_{0.90} = 0.370 \pm 1.746(0.243 \times 0.5)\sqrt{\frac{1}{18} + \frac{(0.5-1.65)^2}{(63.49 - 18 \times 1.65^2)}}$$
$$= (0.288 \rightarrow 0.451)$$

$$\text{At } x_i = 1.5: \quad \langle \mu_{Y|x} \rangle_{0.90} = 0.975 \pm 1.746(0.243 \times 1.5)\sqrt{\frac{1}{18} + \frac{(1.5-1.65)^2}{(63.49 - 18 \times 1.65^2)}}$$
$$= (0.823 \rightarrow 1.127)$$

$$\text{At } x_i = 2.5: \quad \langle \mu_{Y|x} \rangle_{0.90} = 1.670 \pm 1.746(0.243 \times 2.5)\sqrt{\frac{1}{18} + \frac{(2.5-1.65)^2}{(63.49 - 18 \times 1.65^2)}}$$
$$= (1.326 \rightarrow 2.014)$$

$$\text{At } x_i = 3.5: \quad \langle \mu_{Y|x} \rangle_{0.90} = 2.320 \pm 1.746(0.243 \times 3.5)\sqrt{\frac{1}{18} + \frac{(3.5-1.65)^2}{(63.49 - 18 \times 1.65^2)}}$$
$$= (1.512 \rightarrow 3.122)$$

$t_{0.95,16} = 1.746$


Course Flow
Module 1: Probability – Understanding Uncertainty
Foundations of Probability
RVs & Probability Models
Multivariate Probability
Modeling Extremes
Module 2: Statistics – Learning from Data
Descriptive statistics
Statistical Inference
Regression & Statistical Learning
Module 3: Reliability – Decision under Uncertainty
Reliability Analysis
Computational Reliability
Stochastic Processes & Time-Dependent Reliability
Risk-Informed Decision Making

Multiple Linear Reg.
&
Intro to Statistical Learning

Ang and Tang (Section 8.6)

Multiple linear regression

Suppose the dependent variable of interest is Y and that it is a linear function of k random variables $X_1, X_2, \ldots, X_k$. It follows that the mean value of Y at $X_1 = x_{i1}$, $X_2 = x_{i2}, \ldots, X_k = x_{ik}$; $i = 1, 2, \ldots, n$, is

$$y_i' = \beta_o + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} \tag{8.20}$$

for each i, where $\beta_0, \beta_1, \ldots, \beta_k$ are constant regression coefficients that must be estimated from the observed data $(x_{i1}, x_{i2}, \ldots, x_{ik})$, whereas the conditional variance of Y for given values $x_{i1}, x_{i2}, \ldots, x_{ik}$ is assumed to be constant for any i; i.e.,

$$\text{Var}(Y|x_{i1}, x_{i2}, \ldots, x_{ik}) = \sigma^2 \tag{8.21}$$

or is a known function of $(x_{i1}, x_{i2}, \ldots, x_{ik})$; i.e.,

$$\text{Var}(Y|x_{i1}, x_{i2}, \ldots, x_{ik}) = \sigma^2 g^2(x_{i1}, x_{i2}, \ldots, x_{ik}) \tag{8.22}$$

Regression analysis is then performed to estimate $\beta_o, \beta_1, \ldots, \beta_k$ and σ^2 based on a set of observed data of size n, namely, $(x_{i1}, x_{i2}, \ldots, x_{ik}, y_i)$, $i = 1, 2, \ldots, n$.

Multiple linear regression

$$\mathbf{y}' = \mathbf{X}\boldsymbol{\beta} \tag{8.20a}$$

where $\mathbf{y}'$ is a vector $\mathbf{y}' = \{y_1', y_2', \ldots, y_n'\}$, in which each y_i' is given by Eq. 8.20; $\boldsymbol{\beta}$ is a vector of the regression coefficients $\boldsymbol{\beta} = \{\beta_o, \beta_1, \ldots, \beta_k\}$, and $\mathbf{X}$ is an n by $k + 1$ matrix (i.e., n rows and $k + 1$ columns):

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \cdot & \cdot & \cdot & \cdots & \cdot \\ \cdot & \cdot & \cdot & \cdots & \cdot \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix}$$

Let us now concentrate our attention on the case in which the conditional variance is constant. The total squared error for a set of n data points is then

$$\Delta^2 = \sum_{i=1}^{n} (y_i - y_i')^2 = \sum_{i=1}^{n} [y_i - \beta_o - \beta_1 x_{i1} - \cdots - \beta_k x_{ik}]^2 \tag{8.23}$$

Multiple linear regression

Then, on the basis of least squares, we minimize Δ^2 to obtain the following set of linear equations for determining the estimates of β_j, $j = 0, 1, 2, \ldots, k$:

$$\frac{\partial \Delta^2}{\partial \beta_o} = \sum_{i=1}^{n} \left[y_i - \hat{\beta}_o - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik} \right] = 0$$

$$\frac{\partial \Delta^2}{\partial \beta_1} = \sum_{i=1}^{n} x_{i1} \left[y_i - \hat{\beta}_o - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik} \right] = 0 \qquad (8.24)$$

$$\vdots$$

$$\frac{\partial \Delta^2}{\partial \beta_k} = \sum_{i=1}^{n} x_{ik} \left[y_i - \hat{\beta}_o - \hat{\beta}_1 x_{i1} - \cdots - \hat{\beta}_k x_{ik} \right] = 0$$

Equation 8.24 consists of $k + 1$ equations with the $k + 1$ unknown regression coefficients. Again, in matrix notation, the set of equations of Eq. 8.24 may be written as

$$\mathbf{X}^T \mathbf{X} \hat{\beta} = \mathbf{X}^T \mathbf{y} \qquad (8.24a)$$

where $\mathbf{y} = \{y_1, y_2, \ldots, y_n\}$ is the vector of the observed values of Y and T is the transpose.

$$\hat{\beta} = \left(\mathbf{X}^T \mathbf{X} \right)^{-1} \mathbf{X}^T \mathbf{y}$$

Multiple linear regression

Premultiplying both sides of Eq. 8.24a by the inverse of the matrix $\mathbf{X}^T\mathbf{X}$, we obtain the solutions for the least-squares estimates of the regression coefficients as

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} \tag{8.25}$$

where $\hat{\boldsymbol{\beta}}$ is a $(k + 1)$ vector, $\hat{\boldsymbol{\beta}} = \{\beta_o, \beta_1, \ldots, \beta_k\}$, with which we obtain the multiple regression equations, in matrix form,

$$\mathbf{y}' = \mathbf{X}\hat{\boldsymbol{\beta}} \tag{8.26}$$

In scalar form, Eq. 8.26 represents a set of k multiple regression equations as follows:

$$y_i' = \hat{\beta}_o + \sum_{j=1}^{k} \hat{\beta}_j x_{ij}; \quad i = 1, 2, \ldots, n;\ j = 1, 2, \ldots, k \tag{8.26a}$$

where $\hat{\beta}_o$ and $\hat{\beta}_j$ are the components of $\hat{\boldsymbol{\beta}}$.

An unbiased estimate of the conditional variance of Y for given values of $X_1, X_2, \ldots, X_k$ is

$$s^2_{Y|x_1,x_2,\ldots,x_k} = \frac{\Delta^2}{n-k-1} = \frac{\sum_{i=1}^{n}(y_i - y_i')^2}{n-k-1} \tag{8.27}$$

in which y_i' is given by Eq. 8.26a. The reduction in the original variance of Y may also be evaluated with r of Eq. 8.7 by substituting $s_{Y|x_1,\ldots,x_k}$ for $s_{Y|x}$.

Nonlinear Regression

Ang and Tang (Section 8.5)

Thank you!